

1202 labmeeting

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1 previous

교수님 feedback: option 1 모델에서 local latent vector 의 분포에 σ_1^2 추가하고, methodology 정리와 related work 정리하라고 하셨습니다.

2 related works

3 methodology

In this study, we propose a extension of the Latent Space Item Response Model (LSIRM) to accommodate mixed-type data within a unified framework. Specifically, we integrate the LSIRM proposed by Jeon et al. (2021) with the latent space architecture for multilayered networks introduced by Juan and Betancourt (2022). This integration allows us to conceptualize survey data containing heterogeneous variable types as a multilayered structure, where each layer corresponds to a distinct data type. In this section, we first review the foundational concepts of the LSIRM and the latent space model for multilayered networks, followed by the formal derivation of our proposed joint model.

3.1 LSIRM

The Latent Space Item Response Model (LSIRM) is an integrated framework that combines the Rasch model—a foundational approach in Item Response Theory (IRT)—with the latent space model, which captures network structures via a distance-based penalty term. This integration allows for the modeling of higher-order associations between items and respondents.

Traditional Item Response Theory (IRT) models the probability of a specific respondent answering a

specific item correctly based on the item's difficulty and the respondent's ability. Rasch (1961) formalized this perspective using the following model.

$$\text{logit}(\mathbb{P}(Y_{ij} = 1) | \alpha_i, \beta_i) = \alpha_i + \beta_i$$

This formulation models the log-odds of respondent i answering item j correctly as a logistic regression function of the respondent's ability α_i and the item's difficulty β_j . However, this approach has limitations in effectively capturing heterogeneous response patterns where a respondent's performance varies depending on specific item characteristics. Specifically, scenarios where a respondent with high general ability performs poorly in a specific domain cannot be adequately explained using only the ability and difficulty parameters. To address these additional associations between items and respondents, the LSIRM interprets the data as a bipartite network and incorporates concepts from the latent space model. The latent space model explains the formation of edges between nodes using the distance between latent vectors mapped into a latent space. Hoff et al. (2002) described this approach as follows.

$$\text{logit}(\mathbb{P}(Y_{ij} = 1 | \alpha, z_i, z_j)) = \alpha - d(z_i, z_j)$$

This also can be understood as a logistic regression where the probability of an edge forming between nodes i and j is determined by an intercept α and the distance between their latent positions z_i and z_j . The LSIRM adopts this structure to map the higher-order relationships within the bipartite network of item response data—comprising both item and respondent nodes—onto a latent space. Jeon et al. (2021) proposed the LSIRM in the following form.

$$g(\text{Pr}(Y_{ij} = 1)) = \alpha_i + \beta_j - \gamma d(a_i, b_j)$$

Here, α_i and β_j represent ability and difficulty, respectively, consistent with the traditional Rasch model. The term $\gamma d(a_i, b_j)$ captures the higher-order relationships between items and respondents that are not explained by ability and difficulty alone. The parameter γ , which takes a value between 0 and ∞ , serves as a weight that determines the contribution of the interaction term, indicating whether the data is sufficiently explained by the Rasch framework or if the inclusion of higher-order dependencies is necessary. Despite these advancements in modeling network dependencies, a key limitation of the current framework lies in its inability to jointly process heterogeneous data types. As the model operates within a unified Generalized Linear Model (GLM) framework, it imposes a single distributional assumption across the entire dataset. This constraint often necessitates transforming diverse data sources into a uniform format. Such a strategy, however, is problematic as it risks discarding valuable information by obscuring subtle

variations among responses. Furthermore, this mandatory data transformation introduces a potential source of measurement error into the analysis.

3.2 LSM for multilayer network data

Juan and Betancourt (2022) 관점을 서술할 필요가 있을까요? 아래 부분에서 model structure 를 서술하면서 Juan and Betancourt (2022) 를 언급하는데, 따로 chapter 를 할애할 필요가 있을지 모르겠습니다.

3.3 LSIRM with global - local structure

Moving beyond the standard LSIRM perspective that treats item response data as a single bipartite network, we assume that heterogeneous data types reside within a multiplex bipartite network structure. Specifically, let $P^{(1)}$ denote the number of items with binary responses and $P^{(2)}$ denote the number of items with continuous responses. While the responses from n subjects could be represented as a single $n \times (P^{(1)} + P^{(2)})$ bipartite network, we conceptualize them as two distinct layers within a multiplex framework—one of size $n \times P^{(1)}$ and the other of size $n \times P^{(2)}$. Adopting the joint modeling approach proposed by Juan and Betancourt (2022) for such multiplex networks, our model is formulated as follows. The following model description focuses on a two-layered framework handling binary and continuous data. However, note that this framework is readily extensible to scenarios with more than two layers.

binary layer

$$\text{logit}\{Pr(Y_{ij} = 1)\} = \alpha_i^{(1)} + \beta_j^{(1)} - \gamma^{(1)}d(a_i^{(1)}, b_j^{(1)})$$

continuous layer

$$Y_{ij} = \alpha_i^{(2)} + \beta_j^{(2)} - \gamma^{(2)}d(a_i^{(2)}, b_j^{(2)}) + \epsilon$$

such that $\epsilon \stackrel{\text{iid}}{\sim} N(0, \sigma_0^2)$

global structure

$$a_i^{(\ell)} | \sigma_1^2 \stackrel{\text{iid}}{\sim} MVN(z_i, I_d \sigma_1^2)$$

In this framework, the response variables Y_{ij} correspond to distinct data types across different layers. Consequently, we employ layer-specific link functions and distributional assumptions suitable for each data modality. Specifically, the binary layer assumes a Bernoulli distribution with a logit link function, whereas the continuous layer assumes a normal distribution with an identity link function. However, as our model is fundamentally grounded in the Generalized Linear Model (GLM) framework, it offers the flexibility to accommodate various other data types by adopting appropriate link functions and

distributions as needed.

The interpretation of the parameters within the link functions remains consistent with the standard LSIRM framework. Specifically, $\alpha_i^{(\ell)}$ represents the ability (or propensity) of the i -th respondent in layer ℓ , and $\beta_j^{(\ell)}$ denotes the difficulty (or popularity) of the j -th item in layer ℓ . Similarly, $\gamma^{(\ell)}$ acts as a weight parameter for the interaction term, quantifying its significance from a model selection perspective. It specifically measures the extent to which the latent space captures high-order dependencies that the baseline Rasch model fails to explain. Finally, $d(a_i^{(\ell)}, b_j^{(\ell)})$ represents the distance between the latent positions of respondent i and item j within the latent space of layer ℓ .

3.4 Model estimation

We adopt a fully Bayesian approach for model estimation, consistent with the frameworks proposed by Jeon et al. (2021) and Juan and Betancourt (2022). As noted by Hoff et al. (2002), maximum likelihood estimation (MLE) is often intractable in latent space models. This difficulty arises because the likelihood function is non-concave and the model suffers from inherent identifiability issues regarding the latent positions.

3.4.1 prior distribution

We have prior distribution assumption as follows.

No.	Distribution	Index
1	$\alpha_i^{(\ell)} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\alpha^{2,(\ell)})$	$i = 1, \dots, N; \ell = 1, \dots, L$
2	$\sigma_\alpha^{2,(\ell)} \sim IG(a_\sigma^{(\ell)}, b_\sigma^{(\ell)})$	$\ell = 1, \dots, L$
3	$\beta_i^{(\ell)} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \tau_\beta^{2,(\ell)})$	$i = 1, \dots, N; \ell = 1, \dots, L$
4	$\tau_\beta^{2,(\ell)} \sim IG(a_\tau^{(\ell)}, b_\tau^{(\ell)})$	$\ell = 1, \dots, L$
5	$\log(\gamma^{(\ell)}) \sim \mathcal{N}(\mu_\tau^{(\ell)}, \sigma_\tau^{2,(\ell)})$	$\ell = 1, \dots, L$
6	$b_j^{(\ell)} \stackrel{\text{iid}}{\sim} MVN_d(0_d, I_d)$	$j = 1, \dots, P^{(\ell)}; \ell = 1, \dots, L$
7	$z_i \stackrel{\text{iid}}{\sim} MVN_d(0_d, I_d)$	$i = 1, \dots, N$
8	$\sigma_0^2 \sim IG(a_0, b_0)$	—
9	$\sigma_1^2 \sim IG(a_1, b_1)$	—

N is the number of respondent, $P^{(\ell)}$ is the number of item on ℓ th layer and L is the number of layer. d is dimension of latent vectors. Note that all latent vectors are on same dimensional vector space.

We assigned standard normal priors (mean 0, variance 1) to both the item-specific latent vectors and the respondent-specific global latent vectors. This specification restricts the scale of the latent positions, thereby resolving potential identifiability issues.

3.4.2 update rule

The posterior inference is implemented via Markov Chain Monte Carlo (MCMC). We use Metropolis-Hastings (MH) algorithm for parameters without closed-form conditional distributions (e.g., $\alpha, \beta, \gamma, a, b$) and Gibbs sampling for parameters with conjugate priors (e.g., variance terms and z_i). We outline the sampling steps sequentially, starting with the parameters specific to binary layer and continuous layer, followed by the global structure variables.

1. Binary Layer ($\ell = 1$) The posterior distributions of the parameters associated with the binary layer are given by,

$$\begin{aligned} f(\boldsymbol{\alpha}^{(1)}, \boldsymbol{\beta}^{(1)}, \gamma^{(1)}, \mathbf{a}^{(1)}, \mathbf{b}^{(1)} | \mathbf{y}) &\propto \left[\prod_{i=1}^N f(\alpha_i^{(1)}) \right] \times \left[\prod_{j=1}^{P^{(1)}} f(\beta_j^{(1)}) \right] \times f(\gamma^{(1)}) \\ &\times \left[\prod_{i=1}^N f(a_i^{(1)}) \right] \times \left[\prod_{j=1}^{P^{(1)}} f(b_j^{(1)}) \right] \\ &\times \left[\prod_{i=1}^N \prod_{j=1}^{P^{(1)}} P(Y_{i,j} = y_{i,j} | \alpha_i^{(1)}, \beta_j^{(1)}, \gamma^{(1)}, a_i^{(1)}, b_j^{(1)}) \right] \end{aligned}$$

Here, $\boldsymbol{\alpha}^{(1)}, \boldsymbol{\beta}^{(1)}, \mathbf{a}^{(1)}$, and $\mathbf{b}^{(1)}$ denote the sets of all parameters characterizing the first network layer, corresponding to individual parameters $\alpha_i^{(1)}, \beta_j^{(1)}, a_i^{(1)}$, and $b_j^{(1)}$, respectively. This notational convention applies analogously to other layers.

And since $f(\cdot)$ is density function of prior and posterior function and with $\pi_{i,j} = \frac{\exp(\eta_{i,j}^{(1)})}{1 + \exp(\eta_{i,j}^{(1)})}$ where $\eta_{i,j}^{(1)} = \alpha_i^{(1)} + \beta_j^{(1)} - \gamma^{(1)}d(a_i^{(1)}, b_j^{(1)})$, likelihood can be defined as,

$$\mathbb{P}(Y_{i,j} = y_{i,j} | \alpha_i^{(1)}, \beta_j^{(1)}, \gamma^{(1)}, a_i^{(1)}, b_j^{(1)}) = \pi_{i,j}^{y_{i,j}} \times (1 - \pi_{i,j})^{(1-y_{i,j})}$$

1. Update $\alpha_i^{(1)}$ (MH step):

Propose $\alpha_i^{(1)*} \sim \mathcal{N}(\alpha_i^{(1)}, \sigma_{\text{prop alpha}}^2)$. And accept the proposal with probability of

$$\min \left(1, \frac{f(\alpha_i^{(1)*} | \mathbf{y}, \boldsymbol{\alpha}_{-i}^{(1)}, \boldsymbol{\beta}^{(1)}, \gamma^{(1)}, \mathbf{a}^{(1)}, \mathbf{b}^{(1)})}{f(\alpha_i^{(1)} | \mathbf{y}, \boldsymbol{\alpha}_{-i}^{(1)}, \boldsymbol{\beta}^{(1)}, \gamma^{(1)}, \mathbf{a}^{(1)}, \mathbf{b}^{(1)})} \right)$$

such that $\alpha_i^{(1)*}$ is proposed parameter sample and $\boldsymbol{\alpha}_{-i}^{(1)} \stackrel{\text{def}}{=} \boldsymbol{\alpha}^{(1)} \setminus \alpha_i^{(1)}$.

2. **Update $\beta_j^{(1)}$ (MH step):**

Propose $\beta_i^{(1)*} \sim \mathcal{N}(\beta_i^{(1)}, \sigma_{\text{prop beta}}^2)$. And accept the proposal with probability of

$$\min \left(1, \frac{f(\beta_i^{(1)*} | \mathbf{y}, \boldsymbol{\alpha}^{(1)}, \boldsymbol{\beta}_{-i}^{(1)}, \gamma^{(1)}, \mathbf{a}^{(1)}, \mathbf{b}^{(1)})}{f(\beta_i^{(1)} | \mathbf{y}, \boldsymbol{\alpha}^{(1)}, \boldsymbol{\beta}_{-i}^{(1)}, \gamma^{(1)}, \mathbf{a}^{(1)}, \mathbf{b}^{(1)})} \right)$$

such that $\alpha_i^{(1)*}$ is proposed parameter sample and $\beta_{-i}^{(1)} \stackrel{\text{def}}{=} \beta^{(1)} \setminus \beta_i^{(1)}$.

3. **Update $\log \gamma^{(1)}$ (MH step):**

Propose $\log \gamma^{(1)*} \sim \mathcal{N}(\log \gamma^{(1)}, \sigma_{\text{prop}}^2)$. Since $\gamma^{(\ell)}$ has to be non - negative.

Therefore acceptance rate can be transformed from

$$\min \left(1, \frac{p(\gamma^*)q(\gamma|\gamma^*)}{p(\gamma)q(\gamma^*|\gamma)} \right)$$

to

$$\min \left(1, \frac{p(\gamma^*)q(\log(\gamma)|\log(\gamma^*))\gamma}{p(\gamma)q(\log(\gamma^*)|\log(\gamma))\gamma^*} \right)$$

furthermore we can simplify the ratio as follows due to symmetricity of the proposal distribution for $\log \gamma^{(1)}$.

$$\min \left(1, \frac{f(\gamma^{(1)*} | \mathbf{y}, \boldsymbol{\alpha}^{(1)}, \boldsymbol{\beta}^{(1)}, \mathbf{a}^{(1)}, \mathbf{b}^{(1)})\gamma^{(1)}}{f(\gamma^{(1)} | \mathbf{y}, \boldsymbol{\alpha}^{(1)}, \boldsymbol{\beta}^{(1)}, \mathbf{a}^{(1)}, \mathbf{b}^{(1)})\gamma^{(1)*}} \right)$$

4. **Update $a_i^{(1)}$ (MH step):**

Propose $a_i^{(1)*} \sim MVN_d(a_i^{(1)}, \sigma_{\text{prop a}}^2 I_d)$. And accept the proposal with probability of

$$\min \left(1, \frac{f(a_i^{(1)*} | \mathbf{y}, \boldsymbol{\alpha}^{(1)}, \boldsymbol{\beta}^{(1)}, \gamma^{(1)}, \mathbf{a}_{-i}^{(1)}, \mathbf{b}^{(1)})}{f(a_i^{(1)} | \mathbf{y}, \boldsymbol{\alpha}^{(1)}, \boldsymbol{\beta}^{(1)}, \gamma^{(1)}, \mathbf{a}_{-i}^{(1)}, \mathbf{b}^{(1)})} \right)$$

such that $a_i^{(1)*}$ is proposed parameter sample and $\mathbf{a}_{-i}^{(1)} \stackrel{\text{def}}{=} \mathbf{a}^{(1)} \setminus a_i^{(1)}$.

Note that the prior for $a_i^{(1)}$ depends on z_i .

$$p(a_i^{(1)} | z_i) \propto \exp \left(-\frac{1}{2\sigma_1^2} \|a_i^{(1)} - z_i\|^2 \right)$$

5. **Update $b_j^{(1)}$ (MH step):**

Propose $b_i^{(1)*} \sim MVN_d(b_i^{(1)}, \sigma_{\text{prop b}}^2 I_d)$. And accept the proposal with probability of

$$\min \left(1, \frac{f(b_i^{(1)*} | \mathbf{y}, \boldsymbol{\alpha}^{(1)}, \boldsymbol{\beta}^{(1)}, \gamma^{(1)}, \mathbf{a}^{(1)}, \mathbf{b}_{-i}^{(1)})}{f(b_i^{(1)} | \mathbf{y}, \boldsymbol{\alpha}^{(1)}, \boldsymbol{\beta}^{(1)}, \gamma^{(1)}, \mathbf{a}^{(1)}, \mathbf{b}_{-i}^{(1)})} \right)$$

such that $b_i^{(1)*}$ is proposed parameter sample and $\mathbf{b}_{-i}^{(1)} \stackrel{\text{def}}{=} \mathbf{b}^{(1)} \setminus b_i^{(1)}$.

6. Update Variances (Gibbs step):

sample $\sigma_\alpha^{2,(1)}, \tau_\beta^{2,(1)}$ from the full conditional distribution as follows.

$$(a) \sigma_\alpha^{2,(1)} \sim IG(a_\sigma + \frac{N}{2}, b_\sigma + \frac{1}{2} \sum \alpha_i^{(1)2})$$

$$(b) \tau_\beta^{2,(1)} \sim IG(a_\tau + \frac{P_1}{2}, b_\tau + \frac{1}{2} \sum \beta_j^{(1)2})$$

2. Continuous Layer ($\ell = 2$) The posterior distributions of the parameters associated with the continuous layer are given by,

$$\begin{aligned} f(\boldsymbol{\alpha}^{(2)}, \boldsymbol{\beta}^{(2)}, \gamma^{(2)}, \mathbf{a}^{(2)}, \mathbf{b}^{(2)}, \sigma_0^2 | \mathbf{y}) &\propto \left[\prod_{i=1}^N f(\alpha_i^{(2)}) \right] \times \left[\prod_{j=1}^{P^{(2)}} f(\beta_j^{(2)}) \right] \times f(\gamma^{(2)}) \\ &\times \left[\prod_{i=1}^N f(a_i^{(2)}) \right] \times \left[\prod_{j=1}^{P^{(2)}} f(b_j^{(2)}) \right] \times f(\sigma_0^2) \\ &\times \left[\prod_{i=1}^N \prod_{j=1}^{P^{(2)}} P(Y_{i,j} = y_{i,j} | \alpha_i^{(2)}, \beta_j^{(2)}, \gamma^{(2)}, a_i^{(2)}, b_j^{(2)}, \sigma_0^2) \right] \end{aligned}$$

Such that $f(\cdot)$ is density function of prior and posterior function and with $\eta_{ij}^{(2)} = \alpha_i^{(2)} + \beta_j^{(2)} - \gamma^{(2)} d(a_i^{(2)}, b_j^{(2)})$, likelihood can be defined as,

$$\mathbb{P}(Y_{i,j} = y_{i,j} | \alpha_i^{(2)}, \beta_j^{(2)}, \gamma^{(2)}, a_i^{(2)}, b_j^{(2)}, \sigma_0^2) = \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left(-\frac{1}{2\sigma_0^2} (y_{i,j} - \eta_{ij}^{(2)})^2\right)$$

1. Update $\alpha_i^{(2)}, \beta_j^{(2)}, \log \gamma^{(2)}, a_i^{(2)}, b_j^{(2)}$ (MH steps):

These are updated similarly to the binary layer, but the likelihood function is replaced by the Normal density:

$$L(\mathbb{Y} | \cdot) \propto (\sigma_0^2)^{-N/2} \exp\left(-\frac{1}{2\sigma_0^2} \sum_{i,j} (Y_{ij} - \eta_{ij}^{(2)})^2\right)$$

Specifically for $a_i^{(2)}$, the prior is centered at z_i .

2. Update σ_0^2 (Gibbs step):

sample σ_0^2 from the full conditional distribution as follows.

$$\sigma_0^2 | \cdot \sim IG\left(a_0^{(2)} + \frac{N \times P_2}{2}, b_0^{(2)} + \frac{1}{2} \sum_{i,j} (Y_{ij} - \eta_{ij}^{(2)})^2\right)$$

3. Update Variances (Gibbs step):

sample $\sigma_\alpha^{2,(2)}, \tau_\beta^{2,(2)}$ from the full conditional distribution as follows.

$$(a) \sigma_{\alpha}^{2,(2)} \sim IG(a_{\sigma} + \frac{N}{2}, b_{\sigma} + \frac{1}{2} \sum \alpha_i^{(2)2})$$

$$(b) \tau_{\beta}^{2,(2)} \sim IG(a_{\tau} + \frac{P_2}{2}, b_{\tau} + \frac{1}{2} \sum \beta_j^{(2)2})$$

3. Global Structure

1. Update z_i (Gibbs step):

This is a key step linking the two layers. The full conditional distribution of z_i depends on the local positions $a_i^{(1)}$ and $a_i^{(2)}$.

$$p(z_i | a_i^{(1)}, a_i^{(2)}) \propto p(z_i) p(a_i^{(1)} | z_i) p(a_i^{(2)} | z_i) = p(z_i) p(a_i^{(1)}, a_i^{(2)} | z_i)$$

with conditional independence assumption, $p(a_i^{(1)} | z_i, \sigma_1^2) p(a_i^{(2)} | z_i, \sigma_1^2) = p(a_i^{(1)}, a_i^{(2)} | z_i, \sigma_1^2)$

and since it has normal prior, we can use normal - normal conjugacy.

and since $z \sim MVN_p(0, I_p)$ and $a_i^{(\ell)} \sim MVN_p(z_i, I_p \sigma_1^2)$, posterior distribution of z_i is as follows.

$$\Sigma_{z_i} = I \cdot \left(\frac{L}{\sigma_1^2} + 1 \right)^{-1}$$

$$\mu_{z_i} = \Sigma_{z_i} \left(\frac{L \bar{a}_i}{\sigma_1^2} \right)$$

such that,

- L : number of corresponding points for each z_i global latent vectors = number of layers
- $\bar{a}_i$: sample mean of all $a_i^{(\ell)}$ points corresponding to each global latent vector z_i

therefore posterior distribution of z_i follows

$$z_i | a_i^{(1)}, a_i^{(2)} \sim \mathcal{N}(\mu_{z_i}, \Sigma_{z_i})$$

2. Update σ_1^2 (Gibbs step):

let $\mathbf{a}$ is set of all local latent vectors and let $\mathbf{z}$ is set of all global latent vectors. therefore likelihood and prior distribution is as follows.

$$Pr(\mathbf{a} | \mathbf{z}, \sigma_1^2) = (\sigma_1^2)^{-(N*L*d)/2} \exp \left(-\frac{1}{2\sigma_1^2} \sum_{i=1}^N \sum_{\ell=1}^L (a_i^{\ell} - z_i)^2 \right)$$

and,

$$P(\sigma_1^2) = (\sigma_1^2)^{-(a_1+1)} \exp \left(-\frac{b_1}{\sigma_1^2} \right)$$

therefore, posterior distribution of σ_1^2 is as follows.

$$P(\mathbf{a}|\mathbf{z}, \sigma_1^2) = (\sigma_1^2)^{-(a_1 + \frac{N*L*d}{2} + 1)} \exp \left(-\frac{1}{\sigma_1^2} \left[b_1 + \frac{1}{2} \sum_{i=1}^N \sum_{\ell=1}^L (a_i^{(\ell)} - z_i)^2 \right] \right)$$

$$\sigma_1^2 | \mathbf{a}, \mathbf{z} \sim IG \left(a_1 + \frac{n * L * d}{2}, b_1 + \frac{1}{2} \sum_{i=1}^n \sum_{\ell=1}^L (a_i^{(\ell)} - z_i)^2 \right)$$

3.4.3 Identifiability

Due to the inherent non-identifiability of latent space models, it is necessary to align the MCMC samples via Procrustes matching as a post-processing step. This necessity arises because the latent space is invariant to rotation, reflection, and translation, as the coordinate axes do not possess intrinsic interpretations. Consequently, distinct sets of absolute latent coordinates can yield identical likelihoods provided that the relative distances between positions remain preserved. This invariance can obscure MCMC convergence diagnostics; while the relative distances (and thus the model fit) may be stable across iterations, the raw coordinates may drift or rotate, appearing to fail to converge. To mitigate this issue, we employ Procrustes matching to rotate the samples toward a reference configuration (anchor), ensuring consistency in the absolute positions across iterations. We select the sample exhibiting the highest posterior density as this reference configuration. This post-processing technique is a standard practice in the latent space modeling literature. (Hoff et al. (2002), Sewell and Chen (2016), Jeon et al. (2021))

In our proposed framework, each subject is associated with L local latent vectors and one global latent vector, resulting in $L + 1$ sets of latent positions. Crucially, since the local latent positions are hierarchically dependent on the global latent structure, Procrustes matching must be performed jointly on the entire system rather than independently for each layer. To achieve this, we construct a stacked position matrix by concatenating all latent coordinates (both global and local) row-wise for each MCMC iteration. We then apply the Procrustes transformation to this aggregate matrix to maintain the structural integrity of the global-local relationship during the alignment process.

3.5 Model property

아직 어떤 방향으로 서술할지 잘 모르겠어서 놔뒀습니다.

3.5.1 global - local structure

3.5.2 distance metric

References

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